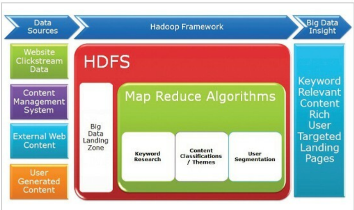

Figure 4: Big Data Hadoop architecture
(Image credits: *googleimages.com*)

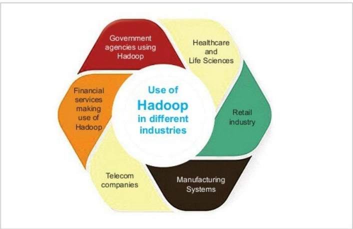

Figure 5: Use of Hadoop in different industries
(Image credits: *googleimages.com*)

their performance is considered as the baseline against which any improvement in the performance observed from presenting the Variation B is measured. Variation A sometimes acts as the original version of the product being tested against what existed before the test. The users exposed to Variation B are called the treatment group.

A/B testing is used to optimise a conversion rate by measuring the performance of the treatment against that of the control group using certain mathematical calculations. This technique takes the guesswork out of website optimisation and enables various data-informed decisions that shift business conversations from what 'we think' to what 'we know'. By measuring the impact that different changes have on our metrics, we can make sure that every change produces positive results.

Association rule learning: Association rule learning is rule-based machine learning used to discover interesting relationships between variables in large databases. It uses a set of techniques for discovering the interesting relationships, also called 'association rules', among different variables present in large databases. All these techniques use a variety of algorithms to generate and test different possible rules. One of its common applications is market basket analysis. This enables a retailer to determine the products frequently bought together and hence use that information for marketing (for example, the discovery that many supermarket shoppers who buy diapers also tend to buy beer). Association rules are being used today in Web usage mining, continuous production, intrusion detection and bioinformatics. These rules do not consider the order of different items either within the same transaction or across different transactions.

Natural language processing: This is a field of computational linguistics and artificial intelligence concerned with the interactions between computers and human languages. It is used to program computers to process large natural language corpora. The major challenges involved in natural language processing (NLP) are natural language generation (frequently from machine-readable logical forms), natural language understanding, connecting the language and machine perception, or some combination thereof. NLP research has relied mostly on machine learning. Initially, many language-processing tasks involved direct hand coding of the rules, which is not suited to natural language variation. Machine-learning pattern calls are now being used instead of statistical inferences to automatically learn different rules through the analysis of large different real-life examples. Many different classes of machine learning algorithms have been applied to NLP tasks. These algorithms take large sets of 'features' as input. These features are generated from the input data.

Research has focused more on the statistical models, which make soft and probabilistic decisions based on attaching the real-value weights to each input feature. The edge that such models have is that they can express the relative certainty of more than one different possible answer rather than only one, hence producing more reliable results as compared to when such a model is included as one of the components of a larger system.

Open source tools to handle Big Data

We have already seen how Big Data is processed and analysed using different techniques. Now, let's go over some of the open source tools that can be used to handle Big Data in order to get some significant value from it.

Apache Hadoop: Apache Hadoop is an open source software framework used for the distributed storage and processing of large data sets using the MapReduce programming model. It consists of computer clusters built using commodity hardware. All the different modules in Hadoop are actually designed with the assumption that different hardware failures are commonly observed occurrences and they should be automatically handled by the framework.

Features:

- The Hadoop framework is mostly written in Java, with some of its native code in C. Its command line utilities are written as shell scripts.
- Apache Hadoop consists of a large storage part, known as the Hadoop Distributed File System.